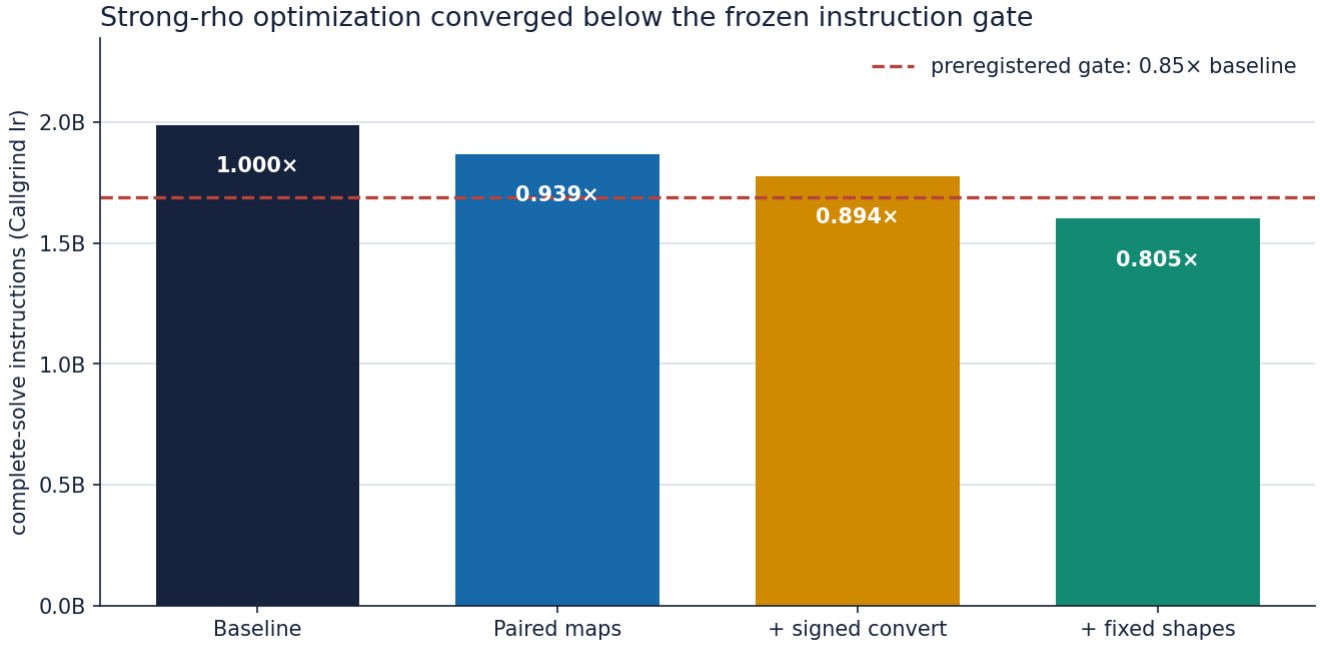
Strong-rho implementation optimization, 2026-10-14

Decision. The standing `rho.signed_frobenius_strong` implementation now uses 0.802155 times the complete-solve Callgrind instructions of its trajectory-identical same-binary legacy path: 1,600,543,701 versus 1,995,304,985 `callgrind.Ir`, a 19.785% reduction. Against the separately archived pre-change binary the ratio is 0.804582. This clears the preregistered 0.85 deterministic gate.

This is an **implementation result**, not a new ECDLP algorithm or generic boundary. Across 240 matched pairs on 48 public targets and six registered Koblitz curves, optimized/legacy charged GAE is exactly 1.0000 with a cluster interval of [1.0000, 1.0000]. All 864 executions verified, all 720 measured runs replayed identically, and the recovered scalars, walk trajectories and every counter stayed fixed.

The native wall gate remains **unknown**. This VM exposes no Linux PSI, so the preregistered L2 run refused before creating a session. A retained L0 diagnostic session was then run with `--allow-busy`, contrary to the protocol's no-fallback clause; its 0.9586 optimized/legacy median and 0.9997 optimized A/A median have no admitted interval and are not performance claims.



n=61 public target W3a517768b166 · one deterministic profile per binary · Valgrind 3.19.0 · session ECBS1hffaa276d8a5c

Frontier boundary

Axis	Position	Evidence class
Cold operations	unchanged: strong rho, `ECBND1h05a2ea918f3c`	canonical
Rho instructions	0.802155 legacy; 0.804582 pre-change	deterministic Callgrind
Native wall	unknown; zero L2 pairs	not admitted
Candidate gaps to rho	unchanged; identical GAE	frozen operations
Historical wall ratios	remain binary-specific	remeasure

The full factor-base, symmetry, F4/F5/F6 and root-finder frontier remains the 2026-10-13 synthesis. The generated index-calculus leaderboard now carries this implementation result as its latest audit, while every ranked row remains unchanged.

What changed

Profile attribution put 1,070,864,152 instructions, 53.91% of the rho-solve interval, under signed-Frobenius canonicalization. The accepted path:

1. applies each normal-basis linear map to both point coordinates in one table traversal;
2. spells out the six-, seven- and eight-byte shapes used by the registered one-word frontier curves;
3. updates both walk coefficients through one modular-reduction dispatch and one shared multiplier conversion; and
4. keeps the literal old path as `rho.signed_frobenius_strong_legacy`, solely for within-binary comparison.

The final canonicalization interval is 678,619,624 instructions, 42.48% of its rho-solve interval. The scalar modular fast path is range-gated below 2^50, falls back to exact `u128` reduction outside that range, and is checked against the prior implementation on randomized residues.

Evidence

Check	Result
Frozen profile	n=61 `W3a517768b166`, public target seed 2026101401, algorithm seed 5480128907060238240
Exact outcome	scalar 41931826107977; 1,591,865 walk steps; 1,698,247 canonicalizations; 1,700,726.570883 GAE
New session	`ECBS1hffaa276d8a5c`, 864/864 verified, 720/720 measured replays identical
Factor-base compatibility	session `ECBS1hb09194fe1d2e`, replay-all 20/20 identical
F4/F5/F6 compatibility	session `ECBS1h2b56c13eeaec`, replay-all 48/48 identical
Root-finder compatibility	session `ECBS1hfd5bf1885227`, eight sampled rho rows identical
Candidate binary	commit `2fd065a6c`; SHA-256 `8778c2a8…31b22` (full value in `BUILD.json`)

Machine-readable values are in `RESULT.json`, the compiler and binary capsule in `BUILD.json`, raw Callgrind parts and hashes in `callgrind/`, and the ECBench records, comparisons, resources and replay receipt in `sessions/run-20261014T012000Z/`. The descriptive run report is `reports/run-20261014T0113Z/`.

Retained attempts

Candidate	Complete-solve Ir	Ratio to archived baseline	Gate
paired coordinate maps and coefficient reduction	1,867,868,681	0.938962	miss
range-proved signed conversion	1,777,918,951	0.893759	miss
fixed six-, seven- and eight-byte shapes	1,600,543,701	0.804582	pass

The first two profiles are preserved rather than dropped.

Scope and graph audit

- Public synthetic targets only; no deployed-key, asymptotic or ECC2K-130 claim.
- `docs/bounds/FRONTIER.md` and `docs/bounds/frontier.json` were checked and remain unchanged because counted operations did not move.
- `docs/index-calculus-scoreboard.html` and `docs/ic/progress-timeline.json` were checked and remain unchanged because this is rho/rho implementation evidence, not a new IC/rho ratio.
- The leaderboard audit was regenerated because every performance handoff is required to appear there.
- Conductor preflight could not run: both `.conductor/WORKFLOW.md` and the `conductor` executable are absent in this checkout.